EEPROM

FT24C32A EEPROM Module

v1.0

2025-09-05

Rev. A

Professional electronic component

PRODUCT OVERVIEW

Professional electronic module designed for reliable performance and easy integration with modern development platforms.

KEY TECHNICAL SPECIFICATIONS

🔌 CONNECTIVITY

Primary Interface:	**GPIO (Interrupt)**
Connector Type:	**JST 4-pin 1.0mm**
Logic Levels:	**VCC-referenced (2V – 5.5V tolerant)**

PIN CONFIGURATION

FUNCTION	NOTES
`GPIO`	Digital I/O

KEY FEATURES

Touch-only sensing
No physical press required – reacts to proximity of a finger.

Auto-calibration
Compensates for environmental changes and drift.

On-board pull-up/down
Ensures clean digital output.

JST PH-2.0 connector
Quick-disconnect cable interface.

Fast response
< 80 ms touch detection time.

Selectable modes
Momentary or toggle output (via solder-jumper on the board).

Mounting holes
Two M3 screw holes for easy panel integration.

🎯 Key Applications
User interfaces for wearables and handheld devices, Touch-activated lamps, buzzers or relays, Capacitive keyboards and remote controls and more

ADDITIONAL TECHNICAL INFORMATION

📦 OVERVIEW

FEATURE	DESCRIPTION
Integrated Chip	FT24C32A 32Kbit I2C EEPROM
Interface	I2C (2-wire)
Operating Voltage	1.8V to 5.5V
Communication Speed	Up to 400 kHz

BASIC COMMANDS

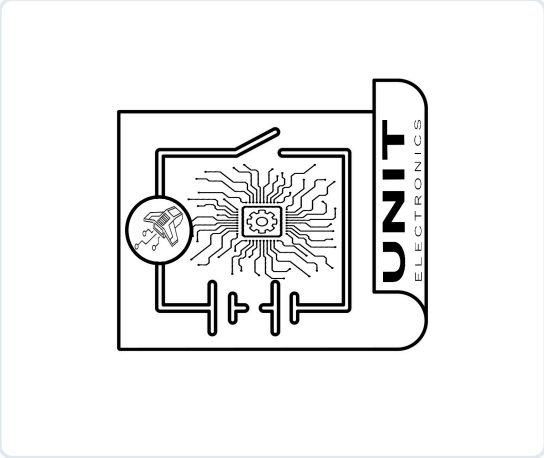
COMMAND	DESCRIPTION	EXAMPLE
`scan`	Scan for I2C devices and detect EEPROM	`scan`
`read `	Read data from memory address	`read 0x0000 16`
`write `	Write text to memory address	`write 0x0000 Hello World`
`erase`	Erase entire EEPROM (fill with 0xFF)	`erase`
`wpcheck [addr]`	Check write protection status	`wpcheck 0x0000`
`addr `	Set EEPROM I2C address manually	`addr 0x50`

EEPROM SPECIFICATIONS

PARAMETER	VALUE
Capacity	32 Kbits (4,096 bytes)
Interface	I2C (2-wire)
I2C Address	0x50 - 0x57 (A0, A1, A2 configurable)
Page Size	32 bytes
Operating Voltage	1.8V - 5.5V
Max Clock Speed	400 kHz
Write Cycle Time	≤ 5ms
Data Retention	> 40 years
Endurance	> 1,000,000 cycles

HARDWARE DOCUMENTATION

CIRCUIT SCHEMATIC



Complete circuit schematic showing all component connections

PIN DESCRIPTION

Detailed pin assignment and electrical specifications

SIGNAL DESCRIPTION

FUNCTION	NOTES
GPIO	Digital I/O

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EEPROM v1.0
Professional Technical Datasheet

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For commercial distribution